

11. Write a C program to find ϵ -closure for all the states in a Non-Deterministic Finite Automata (NFA) with ϵ -moves.

Code:

```
#include <stdio.h>
int a[10][10], v[10], n;
void eclosure(int s)
{
    int i;
    v[s] = 1;
    printf("q%d ", s);
    for(i = 0; i < n; i++)
        if(a[s][i] == 1 && !v[i])
            eclosure(i);
}
int main()
{
    int i, j;
    printf("Enter number of states: ");
    scanf("%d", &n);
    printf("Enter epsilon transition matrix:\n");
    for(i = 0; i < n; i++)
        for(j = 0; j < n; j++)
            scanf("%d", &a[i][j]);
    for(i = 0; i < n; i++)
    {
        for(j = 0; j < n; j++)
            v[j] = 0;
        printf("\nE-Closure(q%d) = { ", i);
        eclosure(i);
        printf("}");
    }
    return 0;
}
```

Output:

```
#include <stdio.h>
int a[10][10], v[10], n;
void eclosure(int s)
{
    int i;
    v[s] = 1;
    printf("q%d ", s);
    for(i = 0; i < n; i++)
        if(a[s][i] == 1 && !v[i])
            eclosure(i);
}
int main()
{
    int i, j;
    printf("Enter number of states: ");
    scanf("%d", &n);
    printf("Enter epsilon transition matrix:\n");
    for(i = 0; i < n; i++)
        for(j = 0; j < n; j++)
            scanf("%d", &a[i][j]);
    for(i = 0; i < n; i++)
    {
        for(j = 0; j < n; j++)
            v[j] = 0;
        printf("\nE-Closure(q%d) = { ", i);
        eclosure(i);
        printf("}");
    }
    return 0;
}
```

```
Enter number of states: 3
Enter epsilon transition matrix:
0 1 0
0 0 1
0 0 0

E-Closure(q0) = { q0 q1 q2 }
E-Closure(q1) = { q1 q2 }
E-Closure(q2) = { q2 }

=== Code Execution Successful ===
```